

Modified Causal Forest: Estimation, Optimal Policy Estimation

Section 1: General information

Welcome to the mcf estimation and optimal policy package.

This report provides you with a summary of specifications and results. More detailed information can be found in the respective output files. Figures and data are provided in the output path in csv-format. This will partly allow you to recreate the figures on your own.

Output path information for MCF ESTIMATION

All outputs: /Users/ghadaelhusseini/PyCharmMiscProject/output

Subdirectories with figures and data are named "ate_iate", "gate", "bgate", and "common support" and contain the content indicated by their name.

Detailed text output: /Users/ghadaelhusseini/PyCharmMiscProject/output/txtFileWithOutput.txt

Summary text output:

/Users/ghadaelhusseini/PyCharmMiscProject/output/txtFileWithOutput_Summary.txt

Output path information for OPTIMAL POLICY ANALYSIS

All outputs: /Users/ghadaelhusseini/PyCharmMiscProject/output1

Detailed text output: /Users/ghadaelhusseini/PyCharmMiscProject/output1/txtFileWithOutput.txt

Summary text output:

/Users/ghadaelhusseini/PyCharmMiscProject/output1/txtFileWithOutput_Summary.txt

BACKGROUND

ESTIMATION OF EFFECTS

The MCF is a comprehensive causal machine learning estimator for the estimation of treatment effects at various levels of granularity, from the average effect at the population level to very fine grained effects at the (almost) individual level. Since most effects at the higher levels are obtained from lower level effects, all effects are internally consistent. Recently, the basic package has been extended to cover new average effects (like Balanced Group Average Treatment Effects, BGATEs, see Bearth and Lechner, 2025, and Quantile Individualized Average Treatment Effects, QIATE, see Kutz and Lechner, 2025) as well as for an optimal policy (decision) module.

Effect estimation is implemented for identification by unconfoundedness and instrumental variables. While unconfoundedness estimation can deal with multiple treatments, instrumental variable estimation is (currently) restricted to binary instruments and binary treatments.

The basis of the MCF estimator is the causal forest suggested by Wager and Athey (2018). Their estimator has been changed in several dimensions which are described in Lechner (2018). The main changes relate to the objective function as well as to the aggregation of effects. Lechner and Mareckova (2025a) provide the MCF's asymptotic guarantees for the case of unconfoundedness. Based on asymptotic theory and a large simulation study, they also compare the MCF to other approaches like the Generalized Random Forest (GRF, Athey, Tibshirani, Wager, 2019) and Double Machine Learning (DML, Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey, Robins, 2018, Knaus, 2022). In this comparison the MCF fared very well, in particular, but not only, for heterogeneity estimation. Some operational issues of the MCF are discussed in Bodory, Busshof, Lechner (2022) and Bodory, Mascolo, and Lechner (2024). There are several empirical studies in different fields using the MCF, like Cockx, Lechner, Boolens (2023), Boller, Lechner, Okasa (2025), or Audrino, Chassot, Huang, Knaus, Lechner, and Ortega (2024), for example.

The asymptotic guarantees (and some finite sample behaviour) of the instrumental variable version of the MCF is discussed in Lechner and Mareckova (2025b).

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References

- Athey, S., J. Tibshirani, S. Wager (2019): Generalized Random Forests, *The Annals of Statistics*, 47, 1148-1178.
 - Athey, S., S. Wager (2019): Estimating Treatment Effects with Causal Forests: An Application, *Observational Studies*, 5, 21-35.
 - Audrino, F., J. Chassot, C. Huang, M. Knaus, M. Lechner, and J.P. Ortega (2024): How does post-earnings announcement sentiment affect firms' dynamics? New evidence from causal machine learning, *Journal of Financial Econometrics*, 22, 1184, 575-604.
 - Bearth, N., and M. Lechner (2025): Causal Machine Learning for Moderation Effects, *Journal of Business & Economic Statistics*.
 - Boller, D., M. Lechner, and G. Okasa (2025): The effect of sport in online dating: evidence from causal machine learning, *Humanities and Social Sciences Communications*, 2025, 12:39.
 - Bodory, H., H. Busshoff, M. Lechner (2023): High Resolution Treatment Effects Estimation: Uncovering Effect Heterogeneities with the Modified Causal Forest, *Entropy*, 24, 1039.
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 - Chernozhukov, V., D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, J. Robins (2018): Double/debiased machine learning for treatment and structural parameters, *Econometrics Journal*, 21, C1-C68.
 - Cockx, B., M. Lechner, J. Bollens (2023): Priority to unemployed immigrants? A causal machine learning evaluation of training in Belgium, *Labour Economics*, 80, Article 102306.
 - Knaus, M. (2022): Double Machine Learning based Program Evaluation under Unconfoundedness, *Econometrics Journal*.
 - Kutz, J, and M. Lechner (2025): Quantile Individualized Average Treatment Effects, *arXiv*.
 - Lechner, M. (2018): Modified Causal Forests for Estimating Heterogeneous Causal Effects, *arXiv*.
 - Lechner, M. (2023): Causal Machine Learning and its Use for Public Policy, *Swiss Journal of Economics & Statistics*, 159:8.
 - Lechner, M., J. Mareckova (2025a): Comprehensive Causal Machine Learning, *arXiv*.
 - Lechner, M., J. Mareckova (2025b): Comprehensive Causal Machine Learning with Instrumental Variables, *arXiv*.
 - Wager, S., S. Athey (2018): Estimation and Inference of Heterogeneous Treatment Effects using Random Forests, *Journal of the American Statistical Association*, 113:523, 1228-1242.
- The optimal policy module offers three (basic) algorithms that can be used to exploit fine grained knowledge about effect heterogeneity to obtain decision rules. The current version is implemented for discrete treatments only.
- There is also an option for different fairness adjustments, as well as an option to take into account estimation risk.

The BEST_POLICY_SCORE algorithm is based on assigning the treatment that has the highest impact at the unit (e.g., individual) level. If the treatment heterogeneity is known (not estimated), this will lead to the best possible result. This algorithm is computationally not burdensome. However, it will not be easy to understand how the implied rules depends on the features of the unit. Its statistical properties are also not clear (for estimated treatment heterogeneity) and there is a certain danger of overfitting, which could lead to an unsatisfactory out-of-training-sample performance.

The BPS_CLASSIFIER classifier algorithm runs a classifier for each of the allocations obtained by the BEST_POLICY_SCORE algorithm. One advantage of this approach compared to the BEST_POLICY_SCORE algorithm is that prediction of the allocation of (new) observations is fast because it does not require to recompute the policy score (as it is the case with the

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BEST_POLICY_SCORE algorithm). The specific classifier is selected among four different classifiers from scikit-learn, namely a simple neural network, two classification random forests with minimum leaf size of 2 and 5, and ADDABOOST. The selection is made according to the out-of-sample performance of the Accuracy Score of scikit-learn.

The POLICY TREE algorithm builds optimal shallow decision trees. While these trees are unlikely to lead to globally optimal allocations, and are computationally much more expensive, they have the advantage that the decision rule is much easier to understand and that some statistical properties are known, at least for certain versions of such decision trees (e.g., Zhou, Athey, Wager, 2023). The basic algorithmic implementation follows the recursive algorithm suggested by Zhou, Athey, Wager (2023) with three (more substantial) deviations (=extensions).

Extension 1: Since using One Hot Encoding for categorical variables may lead to rather extreme leaves for such variables with many different values when building (shallow) trees (splitting one value against the rest), a more sophisticated procedure is used that allows to have several values of the categorical variables on both sides of the split.

Extension 2: Constraints are allowed for. They are handled in a sequential manner: First, an approximate treatment-specific cost vector is obtained and used to adjust the policy score accordingly. Second, trees that violate the constraints are removed (to some extent, optional).

Extensions 3: There are several options implemented to reduce the computational burden, which are discussed below in the section showing the implementation of the policy score.

References

-Zhou, Z., S. Athey, S. Wager (2023): Offline Multi-Action Policy Learning: Generalization and Optimization, Operations Research, INFORMS, 71(1), 148-183.

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Section 2: MCF estimation

METHOD

The standard MCF method used for forest building. Nearest neighbour matching is performed using the Prognostic Score.

Feature selection is not used.

Local centering is used.

Common support is enforced.

VARIABLES

Outcome: outcome

Treatment: treat (with values 0 1 2)

Ordered confounders: x_cont0, x_cont1, x_ord1

Unordered (categorical) confounders: x_unord0

EFFECTS ESTIMATED

Average Treatment Effect (ATE), Individualized Average Treatment Effect (IATE)

NOTE on unordered variables:

One-hot-encoding (dummy variables) is not used as it is expected to perform poorly with trees: It may lead to splits of one category versus all other categories. Instead the approach used is analogous to the one discussed in Chapter 9.2.4 of Hastie, Tibshirani, Friedman (2013), The Elements of Statistical Learning, 2nd edition.

Section 2.1: MCF Training

Training uses 6 CPU cores.

Section 2.1.1: Preparation of training data (mcf training)

METHOD

Variables without variation are removed.

Variables that are perfectly correlated with other variables are removed.

Dummy variables with less than 10 observations in the smaller group are removed.

Rows with any missing values for variables needed for training are removed.

RESULTS

No relevant variables were removed.

Sample size of training data: 1000 (no observations removed).

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Section 2.1.2: Common support (mcf training)

METHOD

The common support analysis is based on checking the overlap in the out-of-sample predictions of the propensity scores (PS) for the different treatment arms. PSs are estimated by random forest classifiers. Overlap is operationalized by computing cut-offs probabilities of the PSs (ignoring the first treatment arm, because probabilities add to 1 over all treatment arms). These cut-offs are subsequently also applied to the data used for predicting the effects.

Overlap is determined by the min / max rule.

Cut-offs for PS are widened by 0.05.

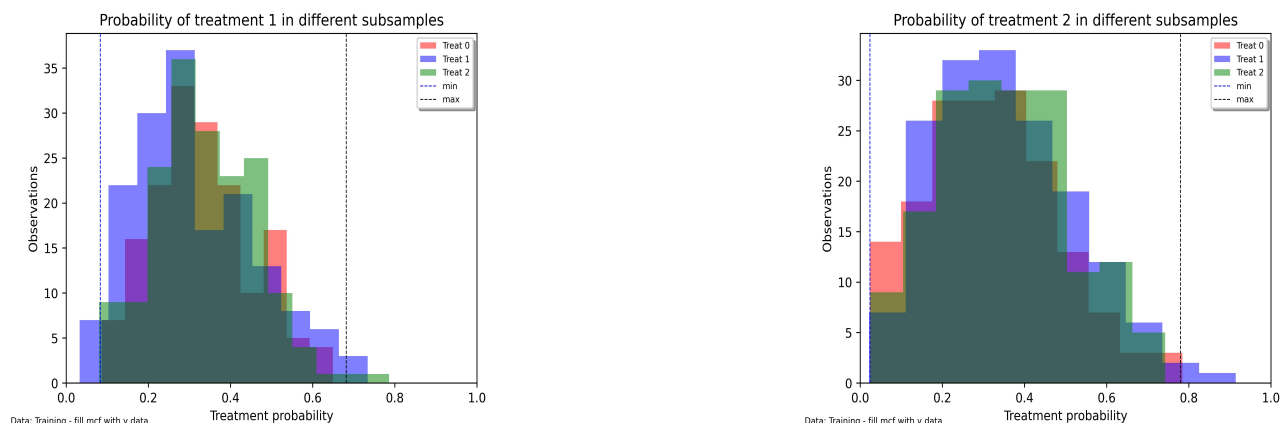
Out-of-sample predictions are generated by 5-fold cross-validation.

RESULTS

Share of observations deleted: 2.20%

Number of observations remaining: 978

Common support plots



Section 2.1.3: Local centering (mcf training)

METHOD

Local centering is based on training a regression to predict the outcome variable conditional on the features (without the treatment). The regression method is selected among various versions of Random Forests, Support Vector Machines, Boosting methods, and Neural Networks of scikit-learn. The best method is selected by minimizing their out-of-sample Mean Squared Error using 5-fold cross-validation. The full set of results of the method selection step are contained in `/Users/ghadaelhusseini/PyCharmMiscProject/output/txtFileWithOutput.txt`.

The respective out-of-sample predictions are subtracted from the observed outcome in the training data used to build the forest. These out-of-sample predictions are generated by 5-fold cross-validation.

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RESULTS

Out-of-sample fit for Random Forest of $E_{y|x}$ (R^2) for outcome: 1.27%

Section 2.1.4: Forest

METHOD and tuning parameters

MSE & MCE Penalty "MSE of treatment variable" is used for forest building.

The causal forest consists of 1000 trees.

The minimum leaf size is 5.

The number of variables considered for each split is 2.

The share of data used in the subsamples for forest building is 67%.

The share of the data used in the subsamples for forest evaluation (outcomes) is 100%.

Alpha regularity is set to 10%.

outcome_lc is the outcome variable used for splitting (locally centered).

The features used for splitting are x_cont0 x_cont1 x_ord1 x_unord0.

CHARACTERISTICS OF THE TRAINED CAUSAL FORESTS

Each tree has on average 43.08 leaves.

Each leaf contains on average 7.6 observations. The median # of observations per leaf is 7.

The smallest leaves have 5 observations.

The largest leaf has 34 observations.

16.45% of the leaves were merged when populating the forest with outcomes from the honesty sample.

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Section 2: MCF estimation

Section 2.2: MCF Prediction of Effects

Training uses 6 CPU cores.

Section 2.2.1: Common support (mcf prediction)

Share of observations deleted: 1.30%

Number of observations remaining: 987

Section 2.2.2: Results

GENERAL REMARKS

The following results for the different parameters are all based on the same causal forests (CF). The combination of the CF with the potentially new data provided leads to weight matrices. These matrices may be large requiring some computational optimisations, such as processing them in batches and saving them in a sparse matrix format. One advantage of this approach is that aggregated effects (ATE, GATE, BGATE) can be computed by aggregation of the weights used for the IATE. Thus a high internal consistency is preserved in the sense that IATEs will aggregate to GATEs, which in turn will aggregate to ATEs.

ESTIMATION

Weights of individual training observations are truncated at 5.00%. Aggregation of IATEs to ATE and GATEs may not be exact due to weight truncation.

INFERENCE

Inference is based on using the weight matrix. Nonparametric regressions are based on k-nearest neighbours.

NOTE

Treatment effects for specific treatment groups (so-called treatment effects on the treated or non-treated) can only be provided if the data provided for prediction contains a treatment variable (which is not required for the other effects).

Section 2.2.2.1: ATE

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RESULT

ATE for outcome

Comparison	Effect	SE	t-value	p-value (%)	Sig.
1 vs 0	0.944	0.172	5.49	0.0	****
2 vs 0	1.015	0.18	5.64	0.0	****
2 vs 1	0.07	0.168	0.42	67.45	

Note: *, **, ***, **** denote significance at the 10%, 5%, 1%, 0.1% level. The results for the potential outcomes can be found in the output files.

Section 2.2.2.2: IATE

This section contains parts of the descriptive analysis of the IATEs. Use the analyse method to obtain more descriptives of the IATEs, like their distribution, and their relations to the features.

RESULTS

Outcome variable: outcome

Comparison	Mean	Median	Std	Effect > 0	mean(SE)	sig 10%	sig 5%	sig 1%
1 vs 0	0.96503	0.72439	0.76687	93.11%	0.40044	56.53%	49.34%	42.05%
2 vs 0	1.01423	0.79889	0.88672	87.23%	0.41083	56.74%	49.44%	43.06%
2 vs 1	0.04920	0.05352	0.35488	55.32%	0.37343	9.32%	4.05%	0.41%

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Section 3: Optimal Policy

METHOD

The assignment rule is based on allocating units to the treatment with the highest score.

VARIABLES provided

Policy scores: y_{pot0} , y_{pot1} , y_{pot2}

Treatment: $treat$

Identifier: id_mcfx

Ordered features: x_cont0 , x_cont1 , x_ord1

Categorical / unordered features: x_unord0

COSTS

No user provided costs of specific treatments.

RESTRICTIONS of treatment shares

Treatment shares are unrestricted.

FAIRNESS

No fairness adjustments performed.

ESTIMATION RISK

No adjustments for estimation error performed.

Section 3.1: Optimal Policy: Training

COMPUTATION

6 logical cores are used for processing.

DATA PREPARATION

Variables without variation are removed.

Variables that are perfectly correlated with other variables are removed.

Dummy variables with less than 10 observations in the smaller group are removed.

Rows with any missing values for variables needed for training are removed.

Section 3.2: Optimal Policy: Evaluation of Allocation(s)

Main evaluation results.

Note: The output files contain relevant additional information, like a descriptive analysis of the treatment groups. Also Qini-like plots are produced and saved in the same location as the text output. Those plots compare the optimal allocation to a reference allocation (3 allocations are used

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as such reference allocations, if available: (i) observed, (ii) random, (iii) the treatment with the highest ATE is allocated to everybody). They show the mean welfare when an increasing share of observations (starting with those who gain most from the optimal allocation compared to the reference allocation) is allocated using the optimal allocation rule.

Treatment allocation (0)

<i>Allocation</i>	<i>Value function</i>	<i>Share of 0 in %</i>	<i>Share of 1 in %</i>	<i>Share of 2 in %</i>
All bb	1.6294	17.1	43.5	39.4
All observed	0.5199	33.3	33.4	33.3
All random	0.5764	30.2	34.0	35.8
All best ATE	0.9264	0.0	100.0	0.0
Switchers bb	1.5779	18.3	42.22	39.48
Switchers random	0.634	29.6	32.82	37.58
Switchers best ATE	0.8974	0.0	100.0	0.0

Note: Allocation analysed is the SAME as the one obtained from the training data (training). bb stands for black box.

Treatment allocation (1)

<i>Allocation</i>	<i>Value function</i>	<i>Share of 0 in %</i>	<i>Share of 1 in %</i>	<i>Share of 2 in %</i>
All bb	1.6897	17.3	42.4	40.3
All observed	0.5886	33.3	33.4	33.3
All random	0.7124	30.2	34.0	35.8
All best ATE	0.9826	0.0	100.0	0.0
Switchers bb	1.7394	18.44	40.33	41.23
Switchers random	0.7291	29.88	33.59	36.53
Switchers best ATE	0.9533	0.0	100.0	0.0

Note: Allocation analysed is the SAME as the one obtained from the training data (prediction). bb stands for black box.